

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

DESIGN AND IMPLEMENTATION OF HUMAN-COMPUTER INTERFACES [NPT-4008]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Which input and output devices would you select for the following systems?
 - A) i) tractor-mounted crop-spraying controller
ii) air traffic control system (5)

For each device, evaluate the alternatives and highlight any relevant reasons why traditional options like the keyboard, mouse, and CRT monitor might be less ideal.
- B) What is the goal and importance of human computer interface. Why human computer interface development is multidisciplinary? Discuss. (5)
- 2) How do you apply the following interaction styles in the interface development of the bus ticket booking system? Explain.
 - a) Menus b) Form fills (5)
- A) Discuss their benefits and drawbacks in HCI.
- B) In a scenario, a single CPU is utilized by various users to execute the tasks they submit via an interface. Which paradigm is most appropriate for this situation? Please elaborate on the chosen paradigm, highlighting its benefits and drawbacks. (5)
- 3) Explain in detail Shneiderman's golden rules for effective design of human computer interface (5)
- A)
- B) Compare and contrast waterfall model and prototyping model for software development. How do you incorporate HCI usability factor in these models? (5)
- 4) Explain the different goals of evaluation. Why it is important to apply evaluation techniques for HCI design? Discuss (5)
- A)
- B) Discuss the different requirements of user support in HCI systems. Give any three examples of user support. (5)
- 5) The developed product must be deployed in the organization for real-time operation. However, organizational issues can break or make a system. Justify this statement with (5)

proper reasons.

A)

B) Suppose a project team has to develop an online meeting platform for group discussions for an organization. How does this new platform affect the communication of the employees in office meetings? (5)

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